

Top Quark Signal in the Cosmological Higgs Collider

1 Top quark propagators and Yukawa vertices

We calculate the Higgs four-point function generated by a top-quark box in exact de Sitter space. The top mass is constant, the chemical potential vanishes, and the external Higgs is a light spectator. We use the Weyl-spinor conventions of Aoki et al. [1], consistent with the corrected free-field formulas of You et al. [2] (v2).

1.1 Higgs field and Majorana decomposition

The metric and scale factor are

$$ds^2 = a^2(\tau)(-d\tau^2 + d\mathbf{x}^2), \quad a(\tau) = -\frac{1}{H\tau}, \quad \tau < 0. \quad (1)$$

We take $\sigma^a = (\mathbf{1}_2, \boldsymbol{\sigma})$ and $\bar{\sigma}^a = (\mathbf{1}_2, -\boldsymbol{\sigma})$. For $\Phi = (0, (h_0 + \delta h)/\sqrt{2})^T$, define

$$g_t = \frac{y_t}{\sqrt{2}}, \quad m_t = g_t h_0, \quad \mu = \frac{m_t}{H}, \quad N_c = 3. \quad (2)$$

The massless Higgs bulk-to-boundary propagator is

$$G_a(k; \tau) = \frac{H^2}{2k^3}(1 - iak\tau)e^{iak\tau}, \quad a = \pm 1. \quad (3)$$

This prescription assumes $|\dot{h}_0| \ll H|h_0|$ and $|m_{h,\text{eff}}^2| \ll H^2$ over the relevant times [5]. The Higgs-to-curvature conversion is not included.

Spinor indices. The spinor indices are raised and lowered by

$$\epsilon_{12} = -\epsilon_{21} = 1, \quad \epsilon^{12} = -\epsilon^{21} = -1, \\ \psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta, \quad \psi_\alpha = \epsilon_{\alpha\beta}\psi^\beta, \quad \epsilon^{\alpha\beta}\epsilon_{\beta\gamma} = \delta^\alpha_\gamma. \quad (4)$$

$$\psi\chi \equiv \psi^\alpha\chi_\alpha = -\epsilon^{\alpha\beta}\psi_\alpha\chi_\beta = \chi\psi, \quad (\psi\psi)^\dagger = \psi_\alpha^\dagger\psi^{\dagger\alpha} = \epsilon^{\dot{\alpha}\dot{\beta}}\psi_\alpha^\dagger\psi_{\dot{\beta}}^\dagger. \quad (5)$$

The symmetry $\psi\chi = \chi\psi$ uses both the Grassmann sign and the antisymmetry of epsilon; Hermitian conjugation reverses operator order.

Let t, t^c be the rescaled left-handed top and the left-handed charge conjugate of the right-handed top, respectively. The latter carries the conjugate gauge representation. The rescaled Dirac-field action is

$$\mathcal{L} = it^\dagger \bar{\sigma}^\mu \partial_\mu t + it^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(m_t + g_t \delta h)(tt^c + \text{h.c.}). \quad (6)$$

With $\tilde{\psi}_I = a^{3/2}\psi_I$, Aoki's field redefinition is

$$t = \frac{\tilde{\psi}_1 + i\tilde{\psi}_2}{\sqrt{2}}, \quad t^c = \frac{\tilde{\psi}_1 - i\tilde{\psi}_2}{\sqrt{2}}. \quad (7)$$

The bilinear $tt^c = t^\alpha(t^c)_\alpha$ is a Lorentz scalar. Its mixed terms cancel under this substitution, giving the coordinate Lagrangian density

$$\mathcal{L} = \sum_{I=1}^2 \left[i\tilde{\psi}_I^\dagger \bar{\sigma}^\mu \partial_\mu \tilde{\psi}_I - \frac{a}{2}(m_t + g_t \delta h)(\tilde{\psi}_I \tilde{\psi}_I + \text{h.c.}) \right]. \quad (8)$$

Gauge interactions are omitted, so the two copies are independent for each color. We calculate one copy and restore $2N_c$ in the box. The factor a in the mass and Yukawa terms is $a^4(a^{-3/2})^2$.

We use $\epsilon_{12} = 1$, $\epsilon^{12} = -1$ and $\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta$, with the same rules for dotted indices. In numerical matrix products,

$$\epsilon = (\epsilon_{\alpha\beta}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma^2, \quad (\epsilon^{\alpha\beta}) = -\epsilon. \quad (9)$$

Every field is Fourier transformed with $e^{i\mathbf{k}\cdot\mathbf{x}}$ and measure $d^3k/(2\pi)^3$. A prime removes the overall momentum delta function.

1.2 Mode functions

For one Majorana copy the helicity spinors obey

$$(\boldsymbol{\sigma} \cdot \hat{\mathbf{k}})h^s = sh^s, \quad P_s = h^s h^{s\dagger} = \frac{\mathbf{1}_2 + s\boldsymbol{\sigma} \cdot \hat{\mathbf{k}}}{2}. \quad (10)$$

With one copy and color suppressed, the operator expansion is

$$\tilde{\psi}_\alpha(\tau, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \sum_{s=\pm} \left[h_\alpha^s u_s b_{\mathbf{k}}^s e^{i\mathbf{k}\cdot\mathbf{x}} + h_{\dot{\alpha}}^{s\dagger} \bar{\sigma}^{0\dot{\alpha}\beta} \epsilon_{\alpha\beta} v_s^* b_{\mathbf{k}}^{s\dagger} e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (11)$$

Here $\{b_{\mathbf{k}}^s, b_{\mathbf{k}'}^{s'\dagger}\} = (2\pi)^3 \delta^{ss'} \delta^3(\mathbf{k} - \mathbf{k}')$. The radial equations are

$$i\partial_\tau u_s + sk u_s = am_t v_s, \quad i\partial_\tau v_s - sk v_s = am_t u_s, \quad (12)$$

$$u_s'' - aH u_s' + (k^2 + a^2 m_t^2 + isaHk)u_s = 0, \quad v_s'' - aH v_s' + (k^2 + a^2 m_t^2 - isaHk)v_s = 0. \quad (13)$$

With $x = -k\tau$, the normalized Bunch–Davies radial modes are

$$u_+ = v_- = \frac{\mu}{\sqrt{2x}} W_{-1/2, i\mu}(-2ix), \quad u_- = v_+ = \frac{i}{\sqrt{2x}} W_{1/2, i\mu}(-2ix). \quad (14)$$

Here u_s, v_s are the two radial functions in the operator expansion (11); the creation term contains v_s^* . Their normalization is $|u_s|^2 + |v_s|^2 = 1$. In particular, $u_+ = v_-$ and $u_- = v_+$, with the relative i displayed above.

Take the boundary value from the BD continuation. At large x the nonvanishing massless-helicity amplitudes tend to $e^{i\pi/4}e^{ix}$, and the other amplitudes are $O(\mu/x)$. The first-order equations conserve $|u_s|^2 + |v_s|^2$, which is one by this boundary condition. Together with $\epsilon P_s^T(-\hat{k})\epsilon^{-1} = P_s(\hat{k})$ this gives the canonical anticommutators, including $\{\tilde{\psi}, \tilde{\psi}\} = 0$.

For the time integrals we use the equivalent Hankel functions

$$F = \frac{\sqrt{\pi x}}{2} e^{\pi\mu/2} H_{1/2-i\mu}^{(1)}(x), \quad G = \frac{\sqrt{\pi x}}{2} e^{-\pi\mu/2} H_{1/2+i\mu}^{(1)}(x). \quad (15)$$

$$u_s = e^{3i\pi/4} \frac{F - sG}{\sqrt{2}}, \quad v_s = e^{3i\pi/4} \frac{F + sG}{\sqrt{2}}, \quad (16)$$

These are scalar radial identities; all propagators below still connect $\tilde{\psi}$ and $\tilde{\psi}^\dagger$.

From the next subsection onward, we suppress the tildes: ψ denotes the conformally rescaled field $\tilde{\psi}$, and D denotes its propagators. The original covariant spinor is no longer used. Thus the propagators below contain no external factors of $a^{-3/2}$; the Yukawa vertices retain the factor $a(\tau)$ in Eq. (8).

1.3 Propagators and vertices

We follow the three propagator types and SK ordering of You et al. [2] (v2), Eqs. (2.30), (2.33), and (2.36), specialized to $\tilde{\lambda} = 0$ and $\tilde{M}_R = \tilde{\nu} = \mu$. Our ψ is their rescaled field $\tilde{N}$; all spinor indices below are lowered using Eq. (4). We denote the two contour branches by $+$, $-$, identified with the numerical values $+1, -1$ in SK vertex weights. There is no overall factor of i in the propagator definitions.

Three types of propagators. With $1 = (\tau_1, \mathbf{k})$ and $2 = (\tau_2, -\mathbf{k})$, define

$$\begin{aligned} D_{ab\alpha\dot{\beta}}(\mathbf{k}; \tau_1, \tau_2) &= \langle T_C \psi_{a,\alpha}(1) \psi_{b,\dot{\beta}}^\dagger(2) \rangle', \\ \widehat{D}_{ab\alpha\beta}(\mathbf{k}; \tau_1, \tau_2) &= \langle T_C \psi_{a,\alpha}(1) \psi_{b,\beta}(2) \rangle', \\ \check{D}_{ab\dot{\alpha}\dot{\beta}}(\mathbf{k}; \tau_1, \tau_2) &= \langle T_C \psi_{a,\dot{\alpha}}^\dagger(1) \psi_{b,\dot{\beta}}^\dagger(2) \rangle'. \end{aligned} \quad (17)$$

Here T_C orders fermions on the closed time contour. The three $-+$ components are the unordered contractions $\langle \psi_1 \psi_2^\dagger \rangle'$, $\langle \psi_1 \psi_2 \rangle'$, and $\langle \psi_1^\dagger \psi_2^\dagger \rangle'$, respectively. Using $u_{sj} = u_s(\tau_j, k)$, $v_{sj} = v_s(\tau_j, k)$ and $P_s = (\mathbf{1}_2 + s\sigma_k)/2$, $\sigma_k = \boldsymbol{\sigma} \cdot \hat{\mathbf{k}}$, their explicit helicity sums are

$$\begin{aligned} D_{-+} &= \sum_{s=\pm 1} u_{s1} u_{s2}^* P_s, \\ \widehat{D}_{-+} &= - \sum_{s=\pm 1} u_{s1} v_{s2}^* P_s \epsilon, \\ \check{D}_{-+} &= \sum_{s=\pm 1} v_{s1} u_{s2}^* \epsilon P_s. \end{aligned} \quad (18)$$

These matrix expressions follow directly from Eq. (11): the creation wavefunction is $v_s^* \epsilon h^{s*}$, and $h^s(\epsilon h^{s*})^T = -P_s \epsilon$. This fixes the epsilon signs when translating the helicity-spinor expressions of Ref. [2] to our lower-index convention.

For the complete special-function expressions, set $x_j = -k\tau_j > 0$ and

$$W_{\pm j} = W_{\pm \frac{1}{2}, i\mu}(-2ix_j), \quad W_{\pm j}^* = W_{\pm \frac{1}{2}, i\mu}(2ix_j). \quad (19)$$

The second equality uses real μ, x_j and $W_{\kappa, \nu} = W_{\kappa, -\nu}$ on the BD boundary values. Substitution of Eq. (14) gives

$$D_{-+}(\mathbf{k}; \tau_1, \tau_2) = \frac{\mu^2 W_{-1} W_{-2}^* (\mathbf{1}_2 + \sigma_k) + W_{+1} W_{+2}^* (\mathbf{1}_2 - \sigma_k)}{4\sqrt{x_1 x_2}}, \quad (20)$$

$$\widehat{D}_{-+}(\mathbf{k}; \tau_1, \tau_2) = \frac{i\mu}{4\sqrt{x_1 x_2}} \left[W_{-1} W_{+2}^* (\mathbf{1}_2 + \sigma_k) - W_{+1} W_{-2}^* (\mathbf{1}_2 - \sigma_k) \right] \epsilon, \quad (21)$$

$$\check{D}_{-+}(\mathbf{k}; \tau_1, \tau_2) = \frac{i\mu}{4\sqrt{x_1 x_2}} \epsilon \left[W_{+1} W_{-2}^* (\mathbf{1}_2 + \sigma_k) - W_{-1} W_{+2}^* (\mathbf{1}_2 - \sigma_k) \right]. \quad (22)$$

The rows and columns carry the indices in Eq. (17). These are exact propagators of the rescaled field and contain no $a^{-3/2}$ endpoint factors.

The four SK components. Fermionic contour ordering gives

$$\begin{aligned} D_{+-\alpha\dot{\beta}} &= -\langle \psi_{\dot{\beta}}^\dagger(2) \psi_\alpha(1) \rangle', \\ \widehat{D}_{+-\alpha\beta} &= -\langle \psi_\beta(2) \psi_\alpha(1) \rangle', \\ \check{D}_{+-\dot{\alpha}\dot{\beta}} &= -\langle \psi_{\dot{\beta}}^\dagger(2) \psi_{\dot{\alpha}}^\dagger(1) \rangle'. \end{aligned} \quad (23)$$

Their explicit forms are

$$\begin{aligned} D_{+-}(\mathbf{k}; \tau_1, \tau_2) &= -\sum_s v_{s2} v_{s1}^* P_{-s}(-\hat{\mathbf{k}}) \\ &= -\frac{W_{+1}^* W_{+2} (\mathbf{1}_2 + \sigma_k) + \mu^2 W_{-1}^* W_{-2} (\mathbf{1}_2 - \sigma_k)}{4\sqrt{x_1 x_2}}, \end{aligned} \quad (24)$$

$$\widehat{D}_{+-}(\mathbf{k}; \tau_1, \tau_2) = \frac{i\mu}{4\sqrt{x_1 x_2}} \left[W_{+1}^* W_{-2} (\mathbf{1}_2 + \sigma_k) - W_{-1}^* W_{+2} (\mathbf{1}_2 - \sigma_k) \right] \epsilon, \quad (25)$$

$$\check{D}_{+-}(\mathbf{k}; \tau_1, \tau_2) = \frac{i\mu}{4\sqrt{x_1 x_2}} \epsilon \left[W_{-1}^* W_{+2} (\mathbf{1}_2 + \sigma_k) - W_{+1}^* W_{-2} (\mathbf{1}_2 - \sigma_k) \right]. \quad (26)$$

Equivalently, the anomalous components obey

$$\mathcal{D}_{+-}(\mathbf{k}; \tau_1, \tau_2) = -\mathcal{D}_{-+}^T(-\mathbf{k}; \tau_2, \tau_1), \quad \mathcal{D} = \widehat{\mathcal{D}}, \check{\mathcal{D}}. \quad (27)$$

The reversed momentum is essential; $P_{-s}(-\hat{\mathbf{k}}) = P_s(\hat{\mathbf{k}})$, as in Eq. (D.46) of Ref. [2]. For each of $\mathcal{D} = D, \widehat{D}, \check{D}$, all four components are

$$\begin{aligned} \mathcal{D}_{-+} &= \langle X(1)Y(2) \rangle', & \mathcal{D}_{+-} &= -\langle Y(2)X(1) \rangle', \\ \mathcal{D}_{++} &= \theta_{12} \mathcal{D}_{-+} + \theta_{21} \mathcal{D}_{+-}, & \mathcal{D}_{--} &= \theta_{21} \mathcal{D}_{-+} + \theta_{12} \mathcal{D}_{+-}, \end{aligned} \quad (28)$$

where $(X, Y) = (\psi, \psi^\dagger), (\psi, \psi)$, or $(\psi^\dagger, \psi^\dagger)$, and $\theta_{ij} = \theta(\tau_i - \tau_j)$. Spinor indices retain the X, Y order in every component.

Late-time expansion. Following Eqs. (2.25)–(2.29) of Ref. [2], let $d = \pm 1$ label the powers $(-2k\tau)^{id\mu}$. At $x = -k\tau \ll 1$, define

$$b_d = e^{-i\pi/4 + \pi d\mu/2} \frac{\mu \Gamma(-2id\mu)}{\Gamma(1 - id\mu)}. \quad (29)$$

The leading radial modes are

$$u_+^L = v_-^L = \sum_{d=\pm 1} b_d (2x)^{id\mu}, \quad u_-^L = v_+^L = \sum_{d=\pm 1} d b_d (2x)^{id\mu}. \quad (30)$$

Each branch has corrections starting at order x . Products of these modes contain terms proportional to $(\tau_1/\tau_2)^{id\mu}$ and terms proportional to $(k^2\tau_1\tau_2)^{id\mu}$. We denote the latter nonlocal branches by (NLoc), d , as in Ref. [2]. A useful zero-chemical-potential form of their coefficient is

$$c_d = \frac{2^{-1-2id\mu}}{\pi} \Gamma(\tfrac{1}{2} - id\mu)^2, \quad c \equiv c_+, \quad c_- = c^*, \quad |c| = \frac{1}{2 \cosh \pi\mu}. \quad (31)$$

The Gamma duplication formula gives the direct normalization map to Eq. (2.28) of Ref. [2]:

$$\frac{c_d}{2} (k^2\tau_1\tau_2)^{id\mu} = \frac{\Gamma(-2id\mu)^2}{\Gamma(-id\mu)^2} (4k^2\tau_1\tau_2)^{id\mu}. \quad (32)$$

In particular, the three radial products are

$$\begin{aligned} [u_{s1} u_{s2}^*]_{(\text{NLoc}),d} &= -\frac{sc_d}{2} (k^2\tau_1\tau_2)^{id\mu}, \\ [u_{s1} v_{s2}^*]_{(\text{NLoc}),d} &= \frac{sdc_d}{2} (k^2\tau_1\tau_2)^{id\mu}, \\ [v_{s1} u_{s2}^*]_{(\text{NLoc}),d} &= -\frac{sdc_d}{2} (k^2\tau_1\tau_2)^{id\mu}. \end{aligned} \quad (33)$$

For a soft propagator momentum $\mathbf{k}_s$, with $k_s = |\mathbf{k}_s|$ and $|k_s\tau_1|, |k_s\tau_2| \ll 1$, the leading and subleading nonanalytic terms are

$$\begin{aligned} [D_{ab\alpha\dot{\beta}}^{\text{LL}}(\mathbf{k}_s; \tau_1, \tau_2)]_{\text{NLoc}} &= [D_{ab\alpha\dot{\beta}}(\mathbf{k}_s; \tau_1, \tau_2)]_{\text{NLoc}} \\ &= - \sum_{d=\pm 1} \frac{2^{-2-2id\mu} \Gamma(\tfrac{1}{2} - id\mu)^2}{\pi} (\tau_1\tau_2)^{id\mu} \left[k_s^{2id\mu} \boldsymbol{\sigma} \cdot \hat{\mathbf{k}}_s + \frac{i(\tau_1 - \tau_2)}{1 + 2id\mu} k_s^{1+2id\mu} \mathbf{1}_2 \right]_{\alpha\dot{\beta}} + \dots, \end{aligned} \quad (34)$$

$$\begin{aligned} [\widehat{D}_{ab\alpha\beta}(\mathbf{k}_s; \tau_1, \tau_2)]_{\text{NLoc}} &= - \sum_{d=\pm 1} \frac{d 2^{-2-2id\mu} \Gamma(\tfrac{1}{2} - id\mu)^2}{\pi} (\tau_1\tau_2)^{id\mu} \left[k_s^{2id\mu} (\boldsymbol{\sigma} \cdot \hat{\mathbf{k}}_s) \boldsymbol{\epsilon} + \frac{i(\tau_1 + \tau_2)}{1 + 2id\mu} k_s^{1+2id\mu} \boldsymbol{\epsilon} \right]_{\alpha\beta} + \dots, \end{aligned} \quad (35)$$

$$\begin{aligned} [\check{D}_{ab\dot{\alpha}\dot{\beta}}(\mathbf{k}_s; \tau_1, \tau_2)]_{\text{NLoc}} &= - \sum_{d=\pm 1} \frac{d 2^{-2-2id\mu} \Gamma(\tfrac{1}{2} - id\mu)^2}{\pi} (\tau_1\tau_2)^{id\mu} \left[k_s^{2id\mu} \boldsymbol{\epsilon} (\boldsymbol{\sigma} \cdot \hat{\mathbf{k}}_s) - \frac{i(\tau_1 + \tau_2)}{1 + 2id\mu} k_s^{1+2id\mu} \boldsymbol{\epsilon} \right]_{\dot{\alpha}\dot{\beta}} + \dots. \end{aligned} \quad (36)$$

The two displayed terms scale as $k_s^{2id\mu}$ and $k_s^{1+2id\mu}$; the omitted terms start at $k_s^{2+2id\mu}$ at fixed endpoint times. Here $\boldsymbol{\epsilon} = i\sigma^2$, and the identity and Pauli matrices carry the displayed spinor indices. These nonlocal terms are the same for all four SK assignments ab , following the cutting prescription of Ref. [2], Eq. (3.13) and Appendix D.3. In the box diagram, these expressions are applied to the two soft-line momenta $\mathbf{q}$ and $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$ separately.

Yukawa vertices. There are two Yukawa vertices, $\delta h\psi\psi$ and $\delta h\psi^\dagger\psi^\dagger$. Their spin tensors, labelled L and $\bar{L}$ respectively, are

$$J_L^{\alpha\beta} = -\epsilon^{\alpha\beta}, \quad J_{\bar{L}}^{\dot{\alpha}\dot{\beta}} = +\epsilon^{\dot{\alpha}\dot{\beta}}, \quad J_L = \epsilon, \quad J_{\bar{L}} = -\epsilon. \quad (37)$$

On SK branch $a = \pm 1$ their amputated rules are $-ia g_t a(\tau) J_r$. The two identical fermion slots cancel the $1/2$ in the interaction. The equivalent calculation with unamputated bilinears and explicit Wick pairings is given in Section 2.2.

2 The four-point box and its soft cut

2.1 Vertex types and the connected loop

We label the two field types by $r, s \in \{L, \bar{L}\}$, where L denotes ψ and $\bar{L}$ denotes $\psi^\dagger$. The propagator used in the following type sums is

$$(D_{ab}^{rs}(\mathbf{k}; \tau_1, \tau_2))_{r,s} = \begin{pmatrix} \widehat{D}_{ab}(\mathbf{k}; \tau_1, \tau_2) & D_{ab}(\mathbf{k}; \tau_1, \tau_2) \\ -D_{ba}^T(-\mathbf{k}; \tau_2, \tau_1) & \check{D}_{ab}(\mathbf{k}; \tau_1, \tau_2) \end{pmatrix}. \quad (38)$$

Each entry is a 2×2 spin matrix with both indices lowered. The lower-left entry follows from fermionic exchange; it is not a fourth independent propagator. Field-type labels r, s , SK labels $a, b = +, -$, and nonlocal branch labels $d = \pm 1$ are distinct.

Each vertex can be of either Yukawa type. For a fixed cycle the 16 choices are grouped as

$(\delta h\psi\psi, \delta h\psi^\dagger\psi^\dagger)$ vertices	$(4, 0)$	$(3, 1)$	$(2, 2)$	$(1, 3)$	$(0, 4)$
number of type assignments	1	4	6	4	1

We keep all of them. For a cycle $i \rightarrow j \rightarrow k \rightarrow l \rightarrow i$ their sum is

$$\mathcal{T}_{ijkl} = \sum_{r_i, r_j, r_k, r_l \in \{L, \bar{L}\}} \text{tr}_2 [J_{r_i} D_{ij}^{r_i r_j} J_{r_j} D_{jk}^{r_j r_k} J_{r_k} D_{kl}^{r_k r_l} J_{r_l} D_{li}^{r_l r_i}]. \quad (39)$$

The all- L term contains four anomalous propagators; mixed vertex assignments contain both normal and anomalous propagators. The matrix contraction is explained below Eq. (40).

2.2 Wick contractions and Majorana counting

The two unamputated interaction insertions are

$$\begin{aligned} & -\frac{ia g_t a(\tau)}{2} \delta h (-\epsilon^{\alpha\beta}) \psi_\alpha \psi_\beta, \\ & -\frac{ia g_t a(\tau)}{2} \delta h \epsilon^{\dot{\alpha}\dot{\beta}} \psi_{\dot{\alpha}}^\dagger \psi_{\dot{\beta}}^\dagger. \end{aligned} \quad (40)$$

For a fixed choice of vertex types around $i \rightarrow j \rightarrow k \rightarrow l \rightarrow i$, write $J_i = J_{r_i}$ and $D_{ij} = D_{ij}^{r_i r_j}$ in this paragraph. Each matrix product contracts one spinor index:

$$(J_i D_{ij})^\alpha{}_\beta = \sum_{\gamma=1}^2 (J_i)^{\alpha\gamma} (D_{ij})_{\gamma\beta}.$$

The indices are dotted or undotted according to the field at each endpoint. Multiplying the four segments and closing the last index with the first gives

$$\text{tr}_2[(J_i D_{ij})(J_j D_{jk})(J_k D_{kl})(J_l D_{li})]. \quad (41)$$

Here tr_2 sums only the two spinor components. The separate sum over the 16 vertex-type choices gives Eq. (39), including both normal and anomalous lines. This contraction assigns no conserved Dirac arrow to a Majorana line. At fourth order expand $\exp[-ig_t \sum_a a \int a(\tau) \delta h(\psi\psi + \text{h.c.})/2]$. The $4!$ attachments of the labelled external Higgs fields cancel $1/4!$. Retain the four bilinear $1/2$ factors for the following count. Of the $7!! = 105$ pairings of eight fermion slots, precisely 48 are connected without self-contractions. They form three undirected cycles, with 2^4 choices of which slot at each vertex connects to its next neighbor. Swapping these slots changes the Wick sign and the antisymmetric J sign together, so all 16 terms for a given cycle are equal. The remaining closed-cycle Wick permutation gives a minus sign. Thus $2^4/2^4 = 1$ for each *undirected* Majorana cycle, or $1/2$ for each of the six written orientations. This count is made before referring to a Dirac trace. A Pfaffian gives the same factor through $\log \text{Pf} = \frac{1}{2} \text{Tr} \log$.

With all external momenta incoming, take the vector routing

$$p_{ij} = q + k_i, \quad p_{jk} = q + k_i + k_j, \quad p_{kl} = q - k_l, \quad p_{li} = q, \quad (42)$$

Each D_{uv} then carries $(p_{uv}; \tau_u, \tau_v; a_u, a_v)$. Write $\mathcal{G} = \langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle'_{\text{conn}}$. For one Majorana copy and one color, and for the full top loop, respectively,

$$\begin{aligned} \mathcal{G}^{(1)} &= -\frac{g_t^4}{2} \sum_{(1,j,k,l)} \sum_{a_1, \dots, a_4 = \pm} \left(\prod_{v=1}^4 a_v \right) \int_{-\infty}^0 \prod_{v=1}^4 [d\tau_v a(\tau_v) G_{a_v}(k_v; \tau_v)] \int \frac{d^3 q}{(2\pi)^3} \mathcal{T}_{1jkl}, \\ \mathcal{G}^{\text{top}} &= 2N_c \mathcal{G}^{(1)}. \end{aligned} \quad (43)$$

The cycle sum runs over permutations of $(j, k, l) = (2, 3, 4)$. The minus sign is the closed-fermion-loop sign. The Majorana factor $1/2$ follows from 48 connected Wick pairings: three undirected cycles with 16 slot assignments each. The external Higgs attachments cancel $1/4!$, as derived above. The 16 vertex types and the six cycle orientations are separate sums.

2.3 The double-soft cut and the cutting rule

The double-soft factorization [4] selects four s -channel cycles as $\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2 \rightarrow 0$: 1234, 1243 near $q \sim k_s$, and 1342, 1432 near $q + k_1 \sim k_s$. The other two cycles have no double-soft s cut for generic hard momenta. In 1234, the soft lines are q and $K = q + k_s$, while the right hard momentum is exactly $q - k_4 = q + k_s + k_3$. Dropping k_s from that line spoils the first nonzero coefficient. Overlapping collapsed channels are not assumed. Applying the nonlocal expansions in Section 1.3 to the field-type components in Eq. (38), the soft cut uses

$$[D_{-+}^{rs}]_{(\text{NLoc})} = [D_{+-}^{rs}]_{(\text{NLoc})}, \quad [D_{ab}^{rs}]_{(\text{NLoc})} = [D_{-+}^{rs}]_{(\text{NLoc})}. \quad (44)$$

The two theta functions therefore add to one on each soft line. This agrees with the corrected cutting prescription of Ref. [2], v2, Sec. 3.2.1 and Appendix D.3. In particular, the normal soft

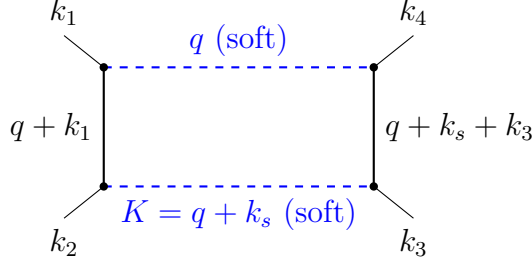


Figure 1: Representative 1234 cycle. All momentum labels are vectors. Dashed lines select the double-soft region; solid internal lines retain complete hard time dependence. Labels refer to the ordered contractions of Eq. (42), not to Majorana charge arrows.

propagator is nonzero and retains σ_k . The hard lines keep their full time dependence and all SK components.

We now evaluate the same-branch assignments $(+, +), (-, -)$, which produce the clock signal, and then the opposite-branch assignments, which produce the nonoscillatory signal.

3 Oscillatory contribution

3.1 Factorized diagram

The four s -cut cycles factorize into two hard subgraphs joined by the soft momentum integral. We include both Majorana copies and all colors from this point, with the normalization in Eq. (43). For $k_s \ll k_L, k_R$, use symmetric hard momenta:

$$\mathbf{k}_1 = \mathbf{k}_L + \frac{1}{2}\mathbf{k}_s, \quad \mathbf{k}_2 = -\mathbf{k}_L + \frac{1}{2}\mathbf{k}_s, \quad (45)$$

$$\mathbf{k}_3 = \mathbf{k}_R - \frac{1}{2}\mathbf{k}_s, \quad \mathbf{k}_4 = -\mathbf{k}_R - \frac{1}{2}\mathbf{k}_s, \quad (46)$$

$$\ell = \mathbf{q} + \frac{1}{2}\mathbf{k}_s, \quad \mathbf{K} = \mathbf{q} + \mathbf{k}_s, \quad \mathbf{n}_L = \hat{\mathbf{k}}_L, \quad \mathbf{n}_R = \hat{\mathbf{k}}_R, \quad \mathbf{n}_s = \hat{\mathbf{k}}_s. \quad (47)$$

Write $k_L = |\mathbf{k}_L|$, $k_R = |\mathbf{k}_R|$; at the accuracy of the final answer $k_L = k_{12}/2$, $k_R = k_{34}/2$, where $k_{ij} = k_i + k_j$ is a sum of magnitudes.

For completeness the four routed cycles, cyclically started at a left hard line, are

cycle	left hard	soft K	right hard	soft q
1234	12 : $k_L + \ell$	23	34 : $k_R + \ell$	41
1243	12 : $k_L + \ell$	24	43 : $-k_R + \ell$	31
1342	21 : $-k_L + \ell$	13	34 : $k_R + \ell$	42
1432	21 : $-k_L + \ell$	14	43 : $-k_R + \ell$	32

All momenta in this table are vectors. The last two rows follow from shifting the original routing variable to $q_{\text{old}} = K - k_1$. It follows directly that their sum is the product of the two independent hard-end ordering sums, after relabelling each pair's dummy times and SK labels. No equality of unequal external magnitudes is assumed here. For the $+i\mu$ branch, the terms retained at the two soft endpoints can be factored as

$$D_{\text{NLoc},+}^{rs}(k; \tau, \tau') \simeq -\frac{c}{2}(k^2 \tau \tau')^{i\mu} L_r^+(k, \tau) \sigma_k U_s^+(k, \tau'). \quad (48)$$

Here $D_{\text{NLoc},+}^{rs}$ denotes the $d = +1$ nonlocal branch, $c = c_+$ is defined in Eq. (31), and $\sigma_k = \boldsymbol{\sigma} \cdot \hat{\mathbf{k}}$. The endpoint matrices through first order are

$$\begin{aligned} L_L^+ &= \mathbf{1}_2 + \frac{i\tau \boldsymbol{\sigma} \cdot \mathbf{k}}{1 + 2i\mu}, & L_L^+ &= \epsilon \left(\mathbf{1}_2 - \frac{i\tau \boldsymbol{\sigma} \cdot \mathbf{k}}{1 + 2i\mu} \right), \\ U_L^+ &= \left(\mathbf{1}_2 + \frac{i\tau \boldsymbol{\sigma} \cdot \mathbf{k}}{1 + 2i\mu} \right) \epsilon, & U_L^+ &= \mathbf{1}_2 - \frac{i\tau \boldsymbol{\sigma} \cdot \mathbf{k}}{1 + 2i\mu}. \end{aligned} \quad (49)$$

The omitted terms at each endpoint are $O(k^2 \tau^2)$. These products reproduce the $d = +1$ terms in Eqs. (34)–(36) through first order. The leading nonzero box signal also requires products of first-order corrections at opposite hard ends.

The left and right hard spin matrices are

$$\begin{aligned} V_L &= -\frac{1}{2} \sum_{r,s=L,\bar{L}} U_r(q, \tau_1) J_r D_{a_1 a_2}^{rs}(\mathbf{k}_L + \boldsymbol{\ell}; \tau_1, \tau_2) J_s L_s(K, \tau_2), \\ V_R &= -\frac{1}{2} \sum_{r,s=L,\bar{L}} U_r(K, \tau_3) J_r D_{a_3 a_4}^{rs}(\mathbf{k}_R + \boldsymbol{\ell}; \tau_3, \tau_4) J_s L_s(q, \tau_4). \end{aligned} \quad (50)$$

Here $L_r = L_r^+$, $U_r = U_r^+$, and the factors $-1/2$ come from the soft lines. Their product reconstructs the type-summed cut as

$$c^2(qK)^{2i\mu} \prod_v (-\tau_v)^{i\mu} \text{tr}_2[V_L \sigma_K V_R \sigma_q]. \quad (51)$$

Let $\mathcal{V}_L, \mathcal{V}_R$ denote the respective time-integrated matrices, with the full SK weights and both Higgs orderings at each end. Strip the factors $H^2/(4k_1^3 k_2^3)$ and $H^2/(4k_3^3 k_4^3)$ from the two subgraphs. The soft-region same-branch contribution then takes the form

$$\mathcal{G}^{\text{top}}|_{s, \text{clock}} = -\frac{N_c g_t^4 H^4}{16 \prod_i k_i^3} \left[c^2 \int \frac{d^3 q}{(2\pi)^3} (qK)^{2i\mu} \text{tr}_2[\mathcal{V}_L \sigma_K \mathcal{V}_R \sigma_q] + \text{c.c.} \right]. \quad (52)$$

The soft time powers $\prod_v (-\tau_v)^{i\mu}$ in Eq. (51) are included in $\mathcal{V}_L, \mathcal{V}_R$.

3.2 Left subgraph and the zeroth-order cancellation

Let $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$ and $\sigma_p = \boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$. Expanding the endpoint matrices in Eq. (49), the four terms in the left type sum are

$$\begin{aligned} 2V_L &= \left(\mathbf{1}_2 + \frac{i\tau_1 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} \right) D^{\text{LL}} \epsilon \left(\mathbf{1}_2 + \frac{i\tau_2 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} \right) \\ &\quad + \left(\mathbf{1}_2 + \frac{i\tau_1 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} \right) D^{\text{L}\bar{L}} \left(\mathbf{1}_2 - \frac{i\tau_2 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} \right) \\ &\quad + \left(\mathbf{1}_2 - \frac{i\tau_1 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} \right) \epsilon D^{\bar{L}L} \epsilon \left(\mathbf{1}_2 + \frac{i\tau_2 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} \right) \\ &\quad + \left(\mathbf{1}_2 - \frac{i\tau_1 \boldsymbol{\sigma} \cdot \mathbf{q}}{1 + 2i\mu} \right) \epsilon D^{\bar{L}\bar{L}} \left(\mathbf{1}_2 - \frac{i\tau_2 \boldsymbol{\sigma} \cdot \mathbf{K}}{1 + 2i\mu} \right) + O(q^2, K^2, qK). \end{aligned} \quad (53)$$

This equation is used through first order in the soft momenta; all four hard propagators carry the arguments in Eq. (50).

After this type sum, only scalar products of the radial functions remain. We denote the three needed scalar time kernels by $\mathcal{W}_{GF}, \mathcal{W}_{FF}, \mathcal{W}_{GG}$:

	$\mathcal{W}_{GF}$	$\mathcal{W}_{FF}$	$\mathcal{W}_{GG}$
$-+$	$G_1 F_2^*$	$F_1 F_2^*$	$G_1 G_2^*$
$+-$	$F_1^* G_2$	$-G_1^* G_2$	$-F_1^* F_2$

Their diagonal SK components follow Eq. (28). In sums such as $\sum_{a,b=\pm} ab$, the signs are the numerical values of the contour labels $+, -$. The labels GF, FF, GG name scalar mode products, not field types. In particular, these kernels are coefficients obtained by performing Eq. (53), not propagators in a rotated spinor basis. For example, at zeroth order its $-+$ component is

$$V_L^{(0),-+} = \frac{1}{2} \sum_s P_s (u_{s1} - v_{s1})(u_{s2}^* + v_{s2}^*) = -G_1 F_2^* \sigma_p. \quad (54)$$

The identity follows by substituting the radial dictionary, with $\sum_s s P_s = \sigma_p$. Thus $V_L^{(0)} = -\mathcal{W}_{GF} \boldsymbol{\sigma} \cdot \mathbf{n}_L$ and $V_R^{(0)} = -\mathcal{W}_{GF} \boldsymbol{\sigma} \cdot \mathbf{n}_R$. At this order $k_1 = k_2 = k_L$, $k_3 = k_4 = k_R$. After relabelling dummy times and SK indices, the two orderings at either end have identical scalar kernels and opposite Pauli matrices. Consequently

$$\sum_{\varepsilon_L, \varepsilon_R = \pm 1} \varepsilon_L \varepsilon_R \text{tr}_2[(\boldsymbol{\sigma} \cdot \mathbf{n}_L) \sigma_K (\boldsymbol{\sigma} \cdot \mathbf{n}_R) \sigma_q] = 0,$$

$\mathcal{G}_{s, \text{clock}}^{\text{box}} \Big|_{k_s^{3 \pm 4i\mu}} = 0.$

(55)

This is an external-leg exchange cancellation. An individual spin trace and an individual hard SK sum need not vanish. The late-time cancellation between SK branches is a separate identity, Eq. (86).

The first nonzero term requires one soft-momentum correction at each hard end: both the hard momentum dependence and the first soft descendants must be retained. A second-order correction at only one end multiplies the vanishing zeroth-order sum at the other. Thus the leading surviving clock term has two extra powers of k_s .

Set $x = -k_L \tau_1$, $y = -k_L \tau_2$, and $r = p/k_L$ for the hard momentum $p = |\mathbf{p}|$. The scalar time integrals are

$$\mathcal{H}_{XY}(\alpha, \beta; r) = \sum_{a,b=\pm} ab \int_0^\infty dx dy x^{\alpha-1} y^{\beta-1} (1 + iax)(1 + iby) e^{-iax - iby} \mathcal{W}_{XY;ab}(r; x, y),$$

$$XY = GF, FF. \quad (56)$$

The mode arguments are rx, ry . We use a common endpoint regulator and combine all four SK components before taking its finite limit. The exact hard-kernel reversal identities are

$$\mathcal{W}_{GF;ab}(\tau_1, \tau_2) = \mathcal{W}_{GF;ba}(\tau_2, \tau_1), \quad \mathcal{W}_{FF;ab}(\tau_1, \tau_2) = -\mathcal{W}_{GG;ba}(\tau_2, \tau_1). \quad (57)$$

Expanding the four field-type contractions gives

$$V_L = -\mathcal{W}_{GF} \sigma_p + \frac{i\tau_1}{1 + 2i\mu} \mathcal{W}_{FF} \boldsymbol{\sigma} \cdot \mathbf{q} - \frac{i\tau_2}{1 + 2i\mu} \mathcal{W}_{GG} \boldsymbol{\sigma} \cdot \mathbf{K} + O(q^2, k_s^2). \quad (58)$$

Hard momentum and endpoint descendants. The first-order hard contraction is Eq. (58). The scalar kernels retain the full hard SK time dependence before expansion of the momentum $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$. The other ordering has $\mathbf{p} = -\mathbf{k}_L + \boldsymbol{\ell}$. The derivative of the hard line and the descendants of both soft endpoints must both be kept. Eq. (57) shows that the first-order difference between the two external magnitudes gives zero in the sum: the integrated $\mathcal{W}_{GF}$ kernel is symmetric in those energies. Also $\int \tau_2 \mathcal{W}_{GG} = -\int \tau_1 \mathcal{W}_{FF}$ at equal external magnitudes.

To evaluate the hard-momentum correction explicitly, write $h(r) = \mathcal{H}_{GF}(i\mu, i\mu; r)$ and $L = \mathbf{n}_L \cdot \boldsymbol{\ell}$. For the ordering $\varepsilon = \pm 1$,

$$|\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}| = k_L + \varepsilon L + O(\ell^2/k_L), \quad (59)$$

$$\boldsymbol{\sigma} \cdot \widehat{\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}} = \varepsilon \boldsymbol{\sigma} \cdot \mathbf{n}_L + \frac{\boldsymbol{\sigma} \cdot (\boldsymbol{\ell} - \mathbf{n}_L L)}{k_L} + O(\ell^2/k_L^2), \quad (60)$$

$$h\left(\frac{|\varepsilon k_L \mathbf{n}_L + \boldsymbol{\ell}|}{k_L}\right) = h(1) + \frac{\varepsilon L}{k_L} h'(1) + O(\ell^2/k_L^2). \quad (61)$$

The $-\mathcal{W}_{GF}\sigma_p$ terms of the two orderings therefore add to

$$-\frac{2}{k_L} \boldsymbol{\sigma} \cdot [h(1)\boldsymbol{\ell} + (h'(1) - h(1))\mathbf{n}_L L] + O(\ell^2/k_L^2). \quad (62)$$

The $\mathcal{W}_{FF}$ and $\mathcal{W}_{GG}$ descendants in Eq. (58) contribute $-4i\mathcal{H}_{FF}(1 + i\mu, i\mu; 1)\boldsymbol{\sigma} \cdot \boldsymbol{\ell}/[k_L(1 + 2i\mu)]$. The common factor $k_L^{-2i\mu}$ comes from $(-\tau_1)^{i\mu}(-\tau_2)^{i\mu}$ after $x = -k_L\tau_1$, $y = -k_L\tau_2$. Finally, the external magnitudes are $|\mathbf{k}_L \pm \mathbf{k}_s/2| = k_L \pm \mathbf{n}_L \cdot \mathbf{k}_s/2 + O(k_s^2/k_L)$. Their first-order shifts cancel under the two orderings by the time-reversal identity in Eq. (57); equal external energies are imposed only after this sum.

Combining these contributions gives

$$\mathcal{V}_L = -2k_L^{-1-2i\mu} \boldsymbol{\sigma} \cdot [\mathcal{P}_1(\mu)\boldsymbol{\ell} + \mathcal{P}_2(\mu)\mathbf{n}_L(\mathbf{n}_L \cdot \boldsymbol{\ell})], \quad (63)$$

$$\mathcal{P}_1(\mu) = \mathcal{H}_{GF}(i\mu, i\mu; 1) + \frac{2i}{1 + 2i\mu} \mathcal{H}_{FF}(1 + i\mu, i\mu; 1), \quad (64)$$

$$\mathcal{P}_2(\mu) = (\partial_r - 1)\mathcal{H}_{GF}(i\mu, i\mu; r)|_{r=1}. \quad (65)$$

The $\mathcal{H}_{FF}$ term comes from the soft descendants; the r derivative acts on the hard line. We evaluate both scalar integrals in Section 3.4.

3.3 Right subgraph

The right subgraph has the same two radial coefficients and its own hard direction:

$$\mathcal{V}_R = -2k_R^{-1-2i\mu} \boldsymbol{\sigma} \cdot [\mathcal{P}_1\boldsymbol{\ell} + \mathcal{P}_2\mathbf{n}_R(\mathbf{n}_R \cdot \boldsymbol{\ell})]. \quad (66)$$

Both $\mathcal{V}_L$ and $\mathcal{V}_R$ already include the two Higgs orderings. Their product in Eq. (52) contains all four s -cut cycles, so no further factor of two or four is applied.

3.4 Scalar time integrals

Relation to the published Hankel scalar seed. The hard time integrals can also be reduced to the scalar seed of Ref. [3], provided its mass index is analytically continued. This gives a direct route to the literature's single- and double-folded formulae. Set

$$\nu_h = \mu - \frac{i}{2}, \quad \rho = i\nu_h = \frac{1}{2} + i\mu, \quad x = -k_L\tau_1, \quad y = -k_L\tau_2, \quad r = p/k_L. \quad (67)$$

Define a dimensionless scalar SK kernel $S_{ab}^{(\nu_h)}$ from the two Wightman functions

$$S_{-+}(r; x, y) = \frac{\pi}{4} e^{-\pi\nu_h} (xy)^{3/2} H_\rho^{(1)}(rx) H_{-\rho}^{(2)}(ry), \quad (68)$$

$$S_{+-}(r; x, y) = \frac{\pi}{4} e^{-\pi\nu_h} (xy)^{3/2} H_{-\rho}^{(2)}(rx) H_\rho^{(1)}(ry), \quad (69)$$

with the standard SK time ordering. Using $H_{-\rho}^{(2)} = e^{-i\pi\rho} H_\rho^{(2)}$ and the mode prefactors in Eq. (15), one finds the *exact* radial identities, for every branch pair,

$$\mathcal{W}_{GF;ab}(r; x, y) = \frac{r}{xy} S_{ab}^{(\nu_h)}(r; x, y), \quad \mathcal{W}_{FF;ab}(r; x, y) = \frac{1}{r} \left(-i\partial_x + \frac{\mu}{x} \right) \mathcal{W}_{GF;ab}(r; x, y). \quad (70)$$

The $1/r$ in the second identity follows from differentiating a mode with argument rx . Differentiating the SK theta functions introduces no delta-function term: the two $\mathcal{W}_{GF}$ Wightman functions agree at $x = y$. Although ν_h is complex, Eq. (69) is an analytic scalar kernel; no physical scalar with complex mass is being introduced.

To see precisely which published seeds are needed, let

$$J_{ab}^{p,q}(r) = \int_0^\infty dx dy x^p y^q e^{-iax - iby} S_{ab}^{(\nu_h)}(r; x, y). \quad (71)$$

In the units $H = 1$ of Ref. [3], its branch-resolved Hankel seed $\mathcal{J}_{ab}^{p,q}$ obeys the exact normalization map

$$J_{ab}^{p,q}(r) = -ab r^{-5-p-q} \mathcal{J}_{ab}^{p,q}(u, u), \quad u = \frac{2r}{1+r}. \quad (72)$$

The reference includes $-ab$ in the seed definition; neither that sign nor the power of r may be omitted. For $p = \alpha - 2$, $q = \beta - 2$, Eq. (56) becomes

$$\mathcal{H}_{GF}(\alpha, \beta; r) = r \sum_{a,b=\pm} ab \left[J_{ab}^{p,q} + ia J_{ab}^{p+1,q} + ib J_{ab}^{p,q+1} - ab J_{ab}^{p+1,q+1} \right]. \quad (73)$$

The four terms correspond to the integer time-power shifts generated by $(1 + ia x)(1 + ib y)$. The SK indices stay explicit until their a, b weights are combined.

The remaining hard coefficient can also be expressed without a new time integral. Put $\mathcal{J}_{ab}^{p,q} = \mathcal{J}_{ab}^{p,q}(1, 1)$, $p_0 = -2 + i\mu + \delta$, and use one common endpoint regulator δ in all seed powers. Equation (73) gives

$$\mathcal{H}_{GF}(i\mu + \delta, i\mu + \delta; 1) = - \sum_{a,b=\pm} \sum_{j,l=0}^1 (ia)^j (ib)^l \mathcal{J}_{ab}^{p_0+j, p_0+l}. \quad (74)$$

Integrating the second relation of Eq. (70) by parts yields

$$\begin{aligned} \mathcal{H}_{FF}(1 + i\mu + \delta, i\mu + \delta; 1) &= i\delta \mathcal{H}_{GF}(i\mu + \delta, i\mu + \delta; 1) \\ &\quad - i \sum_{a,b=\pm} \left[\mathcal{S}_{ab}^{i\mu+\delta,p_0} + ib \mathcal{S}_{ab}^{i\mu+\delta,p_0+1} \right]. \end{aligned} \quad (75)$$

The $i\delta \mathcal{H}_{GF}$ term is retained at finite regulator; the complete $\mathcal{H}_{GF}$ will be finite, so this term vanishes in the final limit. Individual SK terms do have regulator poles. The integration-by-parts boundary term at $x = 0$ cancels only in the complete SK sum, since for $x < y$ the two branches with the same b have the same Wightman function.

The hard-momentum derivative in $\mathcal{P}_2$ follows from the published scalar bootstrap equation without differentiating a time integral. At $u_1 = u_2 = 1$, the coefficient of its second derivative vanishes. Define

$$T_{ab}^{p,q} = \delta_{ab} e^{-ia\pi(p+q)/2} 2^{-5-p-q} \Gamma(5 + p + q), \quad (76)$$

$$\mathcal{D}_{ab}^{p,q} = \frac{1}{2} \sum_{w=p,q} \frac{\nu_h^2 + (w + 5/2)^2}{w + 3} \mathcal{S}_{ab}^{p,q} - \frac{1}{2} \left(\frac{1}{p+3} + \frac{1}{q+3} \right) T_{ab}^{p,q}. \quad (77)$$

Since $du/dr|_{r=1} = 1/2$, $\mathcal{D}_{ab}^{p,q}$ is precisely $d\mathcal{S}_{ab}^{p,q}(u(r), u(r))/dr|_{r=1}$. Combining this with the explicit r^{-5-p-q} in Eq. (72) gives

$$\mathcal{P}_2 = -\text{FP}_{\delta \rightarrow 0} \sum_{a,b=\pm} \sum_{j,l=0}^1 (ia)^j (ib)^l \left[\mathcal{D}_{ab}^{p_0+j,p_0+l} - (5 + 2p_0 + j + l) \mathcal{S}_{ab}^{p_0+j,p_0+l} \right]. \quad (78)$$

Equations (74)–(78), with $\mathcal{P}_1$ from Eq. (65), complete both hard subgraphs in terms of the scalar seed values of Ref. [3]; the right subgraph uses the same two functions.

Regularization and a finite evaluation. There is an essential normalization in Eq. (92) of Ref. [3]: its hypergeometric function is regularized,

$${}_3F_2^{\text{reg}}(a, b, c; d, e; 1) = \frac{{}_3F_2(a, b, c; d, e; 1)}{\Gamma(d)\Gamma(e)}. \quad (79)$$

As an independent normalization check, the conformal scalar $\nu_h = -i/2$ has $S_{-+}(1; x, y) = xy e^{i(x-y)}/2$. Direct elementary time integration gives $\mathcal{S}_{++}^{0,0} = \mathcal{S}_{--}^{0,0} = -3/16$ and $\mathcal{S}_{+-}^{0,0} = \mathcal{S}_{-+}^{0,0} = 1/32$; the implementation reproduces these values and their unequal-power generalization.

The scalar kernel is invariant under $\nu_h \rightarrow -\nu_h$, by the Hankel order-reflection identities. In each equal-SK seed choose the sign of ν_h so that $a i \nu_h = 1/2 + i\mu$. Its ordinary ${}_3F_2(1)$ then has convergence balance $1 + i\mu$, whose real part is positive. This is an exact change of representation and needs no additional mass-index regulator.

For this subsection only set $z = i\mu$, and define the single symmetric family

$$\mathcal{Q}_{jl} = \frac{2^{-1-2z-j-l} \Gamma(1 + 2z + j + l)}{(1 + 2z + j)(1 + 2z + l)} {}_3F_2 \left(\begin{matrix} 1 + 2z + j + l, 1 + z, 1 \\ 2 + 2z + j, 2 + 2z + l \end{matrix}; 1 \right). \quad (80)$$

Only $\mathcal{Q}_{00}, \mathcal{Q}_{01}, \mathcal{Q}_{11}, \mathcal{Q}_{02}, \mathcal{Q}_{12}$ are needed. Also abbreviate the finite endpoint contribution by

$$\mathcal{H}_{\text{end}} = \pi^2 2^{-1-2z} \frac{\Gamma(1 + 2z)^2}{\Gamma(1 + z)^2}. \quad (81)$$

The fully summed hard seeds and momentum derivative are

$$\begin{aligned}\mathcal{H}_{GF}(z, z; 1) &= 2 \cosh(\pi\mu)(\mathcal{Q}_{00} + 2\mathcal{Q}_{01} + \mathcal{Q}_{11}) - \mathcal{H}_{\text{end}}, \\ \mathcal{H}_{FF}(1 + z, z; 1) &= -2i \cosh(\pi\mu)(\mathcal{Q}_{02} + \mathcal{Q}_{12}),\end{aligned}\tag{82}$$

$$\begin{aligned}\mathcal{P}_2 &= 2 \cosh(\pi\mu) \sum_{j,l=0}^1 \left[\frac{j(1+2z+j)}{2(1+z+j)} + \frac{l(1+2z+l)}{2(1+z+l)} - (1+2z+j+l) \right] \mathcal{Q}_{jl} \\ &\quad + (1+2z)\mathcal{H}_{\text{end}} + 2^{-1-2z} \cosh(\pi\mu) \Gamma(1+2z) \frac{4+5z+2z^2}{1+z}.\end{aligned}\tag{83}$$

Together with $\mathcal{P}_1$ in Eq. (65), these are finite, convergent special-function expressions for the complete hard coefficients. No raw mode-power-series truncation or numerical regulator extrapolation is required for the oscillatory signal.

Here is the endpoint cancellation leading to these expressions. For $p_j = -2 + z + \delta + j$, write the Gamma product in the opposite-SK seed as

$$\mathcal{C}_{jl}^\Gamma(\delta) = \frac{\Gamma(\delta+j)\Gamma(1+2z+\delta+j)\Gamma(\delta+l)\Gamma(1+2z+\delta+l)}{\Gamma(1+z+\delta+j)\Gamma(1+z+\delta+l)}.\tag{84}$$

After the external-mode factors $(ia)^j(ib)^l$ and the SK branches are summed, its contribution to $\mathcal{H}_{GF}$ is $2 \cdot 2^{-1-2z-2\delta-j-l} [\cos(\pi\delta) - 1] \mathcal{C}_{jl}^\Gamma(\delta)$. Only $j = l = 0$ has a double pole; therefore

$$\lim_{\delta \rightarrow 0} 2[\cos(\pi\delta) - 1] \mathcal{C}_{00}^\Gamma(\delta) = -\pi^2 \frac{\Gamma(1+2z)^2}{\Gamma(1+z)^2}.\tag{85}$$

This is precisely $-\mathcal{H}_{\text{end}}$ after restoring its power of two. Discarding the poles branch by branch would lose this finite term. The remaining regularized-hypergeometric pieces give $2 \cosh(\pi\mu) \mathcal{Q}_{jl}$. The integration-by-parts insertion in Eq. (75) replaces $j = 0, 1$ by $j = 2$ and contributes $-i$ times the same pieces; it has no double-pole remainder. Finally, substituting p_j in Eq. (77) gives the bracket in Eq. (83). The last term in that equation is the summed bootstrap source, and $(1+2z)\mathcal{H}_{\text{end}}$ is the derivative's endpoint remainder. These steps preserve all factors of r .

The late-time cancellation can also be seen before integration. For either scalar kernel $\mathcal{W}$, with $\mathcal{K}_a(x) = (1 + iax)e^{-iax}$, the SK sum for $x < y$ is

$$\sum_{a,b} ab \mathcal{K}_a(x) \mathcal{K}_b(y) \mathcal{W}_{ab}(x, y) = [\mathcal{K}_+(x) - \mathcal{K}_-(x)][\mathcal{K}_+(y) \mathcal{W}_{-+}(x, y) - \mathcal{K}_-(y) \mathcal{W}_{+-}(x, y)].\tag{86}$$

Since $\mathcal{K}_+(x) - \mathcal{K}_-(x) = -2ix^3/3 + O(x^5)$, the common late-time endpoint is integrable in the hard blocks used here. The early-time BD prescription is still necessary; this identity does not justify integrating undamped oscillations numerically on a truncated interval.

For reference, direct evaluation of Eqs. (82) and (83) gives

μ	$\mathcal{P}_1$	$\mathcal{P}_2$
0.20	$-1.030411081 - 1.109020094i$	$3.163771501 + 0.704629904i$
0.73	$-1.598130708 - 2.346219694i$	$1.923885009 + 1.958596831i$
1.00	$-1.325125325 - 2.665004109i$	$1.137292037 + 2.131408832i$
2.00	$1.901463897 - 2.535666708i$	$-1.119241948 + 1.383844867i$
5.00	$3.921820606 + 1.458261456i$	$-2.047325876 - 0.690483918i$

The branch-resolved calculation converges to these values with a common endpoint regulator. The two representations are checked in `checks/weyl_seed_checks.py`; the physical coefficients are implemented in `checks/weyl_scalar_seed.py`.

3.5 Central momentum integral

Choose the angle convention

$$\mathbf{n}_L = (\sin \theta_1, 0, \cos \theta_1), \quad \mathbf{n}_R = (\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3), \quad \mathbf{n}_s = (0, 0, 1). \quad (87)$$

$$\text{tr}[(\boldsymbol{\sigma} \cdot \mathbf{A})\sigma_K(\boldsymbol{\sigma} \cdot \mathbf{B})\sigma_q] = 2[(\mathbf{A} \cdot \hat{K})(\mathbf{B} \cdot \hat{q}) + (\mathbf{A} \cdot \hat{q})(\mathbf{B} \cdot \hat{K}) - (\mathbf{A} \cdot \mathbf{B})(\hat{K} \cdot \hat{q})]. \quad (88)$$

The spin trace of Eq. (88) leaves a polynomial numerator divided by qK . To display it, write the two vectors appearing in the hard blocks as

$$\mathbf{v}_L = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_L (\mathbf{n}_L \cdot \boldsymbol{\ell}), \quad \mathbf{v}_R = \mathcal{P}_1 \boldsymbol{\ell} + \mathcal{P}_2 \mathbf{n}_R (\mathbf{n}_R \cdot \boldsymbol{\ell}). \quad (89)$$

Since $\mathbf{K} = \boldsymbol{\ell} + \mathbf{k}_s/2$ and $\mathbf{q} = \boldsymbol{\ell} - \mathbf{k}_s/2$, the scalar loop integral after the spin contraction is

$$2 \int \frac{d^3 q}{(2\pi)^3} (qK)^{2i\mu-1} \left[2(\mathbf{v}_L \cdot \boldsymbol{\ell})(\mathbf{v}_R \cdot \boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L \cdot \mathbf{k}_s)(\mathbf{v}_R \cdot \mathbf{k}_s) - (\mathbf{v}_L \cdot \mathbf{v}_R)(\ell^2 - k_s^2/4) \right]. \quad (90)$$

Its mass dimension is $5 + 4i\mu$. We evaluate it using the scalar convolution

$$F_0(A, B) = \frac{\Gamma(A+B-\frac{3}{2})\Gamma(\frac{3}{2}-A)\Gamma(\frac{3}{2}-B)}{\Gamma(A)\Gamma(B)\Gamma(3-A-B)}, \quad (91)$$

$$\int \frac{d^3 q}{(2\pi)^3} (q^2)^{-A} (K^2)^{-B} = \frac{k_s^{3-2A-2B}}{(4\pi)^{3/2}} F_0(A, B). \quad (92)$$

For the scalar numerator this identity follows by Schwinger parameters and a Gaussian momentum integral in the convergence domain $\Re A, \Re B < 3/2$, $\Re(A+B) > 3/2$. The shifted polynomial moments are defined by the same meromorphic continuation, with common regulator parameters before recombination. The physical soft-region integrals have UV power divergences when extended beyond $q \sim k_s$; this continuation selects their nonanalytic coefficient. UV-dependent polynomial contact terms belong to matching with the hard region and are excluded. No physical UV cutoff is sent to infinity term by term. The following reduction expresses the polynomial numerator as a finite sum of these scalar convolutions. The result depends on the two hard directions relative to $\mathbf{k}_s$ as well as on their mutual angle.

Reduction of the polynomial numerator. For the reduction only, measure q, K, ℓ in units of k_s , put $u = q^2$, $v = K^2$, and set $\nu = 1/2 - i\mu$. Then

$$\ell^2 = (u + v)/2 - 1/4, \quad \ell_{\parallel} \equiv \boldsymbol{\ell} \cdot \mathbf{n}_s = (v - u)/2, \quad \ell_{\perp}^2 = \ell^2 - \ell_{\parallel}^2. \quad (93)$$

Azimuthal averaging replaces two transverse components by $\ell_{\perp}^2 \delta_{ij}^{\perp}/2$ and four by $\ell_{\perp}^4 (\delta_{ij}^{\perp} \delta_{kl}^{\perp} + \delta_{ik}^{\perp} \delta_{jl}^{\perp} + \delta_{il}^{\perp} \delta_{jk}^{\perp})/8$, where $\delta_{ij}^{\perp} = \delta_{ij} - n_{si} n_{sj}$. The numerator in Eq. (90) consequently becomes a polynomial

in u, v and the three external angles. Each monomial is integrated by

$$u^m v^n \longmapsto \frac{F_0(\nu - m, \nu - n)}{(4\pi)^{3/2}}, \quad (94)$$

$$\frac{F_0(\nu - m, \nu - n)}{F_0(\nu, \nu)} = \frac{(3/2 - \nu)_m (3/2 - \nu)_n (1 - \nu)_m (1 - \nu)_n}{(5/2 - 2\nu)_{m+n} (3 - 2\nu)_{m+n}}. \quad (95)$$

Only nonnegative integers m, n with $m + n \leq 4$ occur. Thus the bubble is a finite Gamma sum, and all its terms reduce to the same $F_0(\nu, \nu)$ times rational functions. For example the coefficient of $\mathcal{P}_1^2$ in the square bracket of Eq. (90) is $\ell^4 + \ell^2/4 - \ell_{\parallel}^2/2$, whose integral is

$$\frac{F_0(\nu, \nu)}{(4\pi)^{3/2}} \frac{3(\nu - 1)(2\nu - 3)}{8(\nu - 2)(4\nu - 7)(4\nu - 5)}. \quad (96)$$

The $\mathcal{P}_1 \mathcal{P}_2$ and $\mathcal{P}_2^2$ terms are evaluated by the same two steps. Their fully reduced coefficients are displayed directly in Eq. (103), without introducing separate symbols for each tensor moment. The azimuthal average introduces only the five angle monomials 1, $c_1^2 + c_3^2$, $c_1^2 c_3^2$, $\sin(2\theta_1) \sin(2\theta_3) \cos \phi_3$, and $\sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3)$, where $c_i = \cos \theta_i$. Applying Eq. (95) to their coefficients reproduces Eq. (103).

3.6 Total oscillatory trispectrum

The leading nonzero oscillatory term, summed over the four s -cut boxes, is

$$\begin{aligned} \langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle' |_{s, \text{clock}} = & - \frac{N_c g_t^4 H^4 k_s^5}{16 k_1^3 k_2^3 k_3^3 k_4^3 k_L k_R} \\ & \times \left[\left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\mu} \mathcal{C}(\mu; \theta_1, \theta_3, \phi_3) + \text{c.c.} \right]. \end{aligned} \quad (97)$$

The angular dependence can be written as

$$\begin{aligned} \mathcal{C} = & A_1 + A_2(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3 \cos^2 \theta_1 \cos^2 \theta_3 \\ & + B_1 \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 + D_1 \sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3). \end{aligned} \quad (98)$$

All five coefficients share one soft-bubble normalization, including the two soft-mode coefficients c^2 , the factor 4 from the two Bose-summed hard blocks, the factor 2 from the Pauli trace, and the phase from the chosen momentum ratio:

$$\mathcal{B}(\mu) = \frac{2^{1+4i\mu} \Gamma(\frac{1}{2} - i\mu)^4 F_0(\nu, \nu)}{32\pi^2 (4\pi)^{3/2} (\nu - 3)(\nu - 2)(4\nu - 7)(4\nu - 5)}, \quad \nu = \frac{1}{2} - i\mu. \quad (99)$$

In particular, the natural hard scaling is $(k_s^2/(k_L k_R))^{2i\mu}$. Since $4k_{12}k_{34} = 16k_L k_R$ at this order, using the displayed ratio transfers $16^{2i\mu}$ into the coefficient. Thus $8c^2 16^{2i\mu} = 2^{1+4i\mu} \Gamma(1/2 - i\mu)^4 / \pi^2$, which fixes both the phase and the overall factor in Eq. (99). Combining the bubble with the two hard coefficients of Eq. (65) gives the final mass dependence:

$$A_1 = \mathcal{B}(\nu - 1)(2\nu - 3) [4(\nu - 3)(3\mathcal{P}_1^2 + 2\mathcal{P}_1 \mathcal{P}_2) + (2\nu - 5)\mathcal{P}_2^2], \quad (100)$$

$$A_2 = -\mathcal{B}\mathcal{P}_2^2(\nu - 1)(2\nu - 3)^2, \quad A_3 = -3A_2, \quad (101)$$

$$B_1 = -2\mathcal{B}\mathcal{P}_2^2(\nu - 1)(\nu - 2)(2\nu - 3), \quad (102)$$

$$D_1 = \mathcal{B}\mathcal{P}_2^2(\nu - 2)(2\nu - 5)(2\nu - 3). \quad (103)$$

Thus the answer requires only two hard functions $\mathcal{P}_1, \mathcal{P}_2$ and one explicit bubble factor $\mathcal{B}$. The coefficients are complex and include the clock phase. The full answer is even under either Higgs-pair exchange and under parity. Its engineering dimension is -5 , as required for the primed four-point function. The powers of H follow from four rescaled Weyl Yukawa vertices and four external Higgs contractions; no extra factor of H or a second color/orientation multiplicity is implicit. For fixed $\mu > 0$ these are the leading double-soft nonanalytic terms; higher soft powers and analytic contact terms are not displayed. With the centered hard momenta the next clock terms begin at $k_s^{7\pm 4i\mu}$, and the next mixed terms at k_s^5 . Thus the generic relative error within each branch is $O(k_s^2/k_L^2, k_s^2/k_R^2, k_s^2/(k_L k_R))$; it is not a relative-error bound at a zero of the displayed angular coefficient.

4 Nonoscillatory contribution

4.1 Left and right subgraphs

Taking opposite soft branches gives a nonoscillatory k_s^3 term. For $d = -1$, the leading terms in Eqs. (34)–(36) are represented by

$$L_L^- = \mathbf{1}_2, \quad L_L^- = -\epsilon, \quad U_L^- = \epsilon, \quad U_L^- = -\mathbf{1}_2,$$

with scalar prefactor $+c^*/2$, compared with $-c/2$ for $d = +1$. Use U_r^- at one soft endpoint and L_s^+ at the other in Eq. (50), and interchange $+, -$ at the right end. The superscripts $-+$ and $+-$ below label the two nonlocal mass branches at the endpoints; the SK labels remain on the hard propagator. At zeroth order their explicit four-type sums give

$$\begin{aligned} V_L^{-+, (0)} &= -\frac{1}{2} \sum_{r,s} U_r^- J_r D^{rs} J_s L_s^+ = \mathcal{W}_{FF} \mathbf{1}_2, \\ V_R^{+-, (0)} &= -\frac{1}{2} \sum_{r,s} U_r^+ J_r D^{rs} J_s L_s^- = -\mathcal{W}_{GG} \mathbf{1}_2. \end{aligned} \tag{104}$$

The soft prefactor for this assignment is $-|c|^2 (q^2)^{-i\mu} (K^2)^{i\mu}$, from the opposite signs of the two branch prefactors displayed above. Its minus sign cancels the closed-loop sign in Eq. (43). The endpoint time powers are included when integrating these hard matrices. Equation (57) makes both integrated hard coefficients equal to the real seed $\mathcal{H}_{FF}(-i\mu, +i\mu; 1)$. Each hard end therefore has a Bose-even zeroth-order term. There are two soft assignments and two Higgs orderings at each end, giving $2 \times 2 \times 2$ contributions; the spin trace adds $2\hat{q} \cdot \hat{K}$. The soft coefficients give $|c|^2 = 1/(4 \cosh^2 \pi\mu)$. These factors fix the normalization of the mixed term independently of the oscillatory branch.

4.2 The mixed hard coefficient

For evaluation of the remaining mixed seed, the same radial identity and integration by parts give the general formula

$$\begin{aligned} \mathcal{H}_{FF}(\alpha, \beta; 1) &= [\mathrm{i}(\alpha - 1) + \mu] \mathcal{H}_{GF}(\alpha - 1, \beta; 1) \\ &\quad - \mathrm{i} \sum_{a,b=\pm} \left[\mathcal{S}_{ab}^{\alpha-1, \beta-2} + \mathrm{i}b \mathcal{S}_{ab}^{\alpha-1, \beta-1} \right]. \end{aligned} \quad (105)$$

Set $\alpha = -\mathrm{i}\mu + \delta$, $\beta = \mathrm{i}\mu + \delta$, perform the complete sum, and then take $\delta \rightarrow 0$ using the corrected regularized seeds. The conjugation relation $\mathcal{H}_{FF}(\alpha, \beta; 1)^* = \mathcal{H}_{FF}(\beta^*, \alpha^*; 1)$ proves the mixed seed is real. The code evaluates this limit by fourth-order extrapolation, with step-size and precision checks; the clock coefficients above instead use the analytic limit. For example, $\mathcal{H}_{FF}(-0.73\mathrm{i}, +0.73\mathrm{i}; 1) = -1.388500280198814$; halving the extrapolation step changes this value by less than 2×10^{-15} .

4.3 Central integral and total trispectrum

The scalar bubble in this case is

$$\int \frac{\mathrm{d}^3 q}{(2\pi)^3} (q^2)^{-\mathrm{i}\mu} (K^2)^{\mathrm{i}\mu} \hat{q} \cdot \hat{K} = \frac{k_s^3}{(4\pi)^{3/2}} \frac{2\sqrt{\pi}}{3} \mu(1 + \mu^2) \coth(\pi\mu). \quad (106)$$

After both soft branches and all four s -cut boxes are summed, the result is

$$\langle \delta h^4 \rangle' \Big|_{s, \text{mixed}} = \frac{N_c g_t^4 H^4 k_s^3}{16 \prod_i k_i^3} \frac{2\mu(1 + \mu^2)}{3\pi \sinh(2\pi\mu)} \left[\mathcal{H}_{FF}(-\mathrm{i}\mu, +\mathrm{i}\mu; 1) \right]^2. \quad (107)$$

It is independent of angles at this order. Equation (107) is the leading nonoscillatory nonanalytic term, whereas Eq. (97) is the leading mass-dependent clock signal. At $\mu = 0$ the separated branches coalesce; the limit must be taken after combining the relevant terms. Here $k_s^3 = (\mathbf{k}_s^2)^{3/2}$ is nonanalytic as a function of the soft momentum components, even though it has no mass-dependent oscillation. It cannot be discarded as a polynomial contact term. The remaining real coefficient is evaluated with the same scalar seed through Eq. (105). For example, $\mathcal{H}_{FF}(-0.73\mathrm{i}, +0.73\mathrm{i}; 1) = -1.388500280198814$.

The results above include the leading oscillatory and nonoscillatory nonanalytic terms for a single collapsed channel. Analytic contact terms, overlapping collapsed channels and Higgs-to-curvature conversion are outside the calculation. Section 5 specifies the numerical checks and their limits.

5 Numerical checks and scope

The Whittaker implementation constructs all normal and anomalous contractions directly from Eq. (11). It verifies EOMs, normalization and the Hankel map at $\mu = 0.2, 0.73, 2, 5$. The explicit Whittaker products in Section 1.3 are checked separately in `checks/explicit_propagator_checks.py`: all three propagators and all SK assignments agree with the mode contractions at four masses, three momentum directions and three time pairs. An alternative Hankel

calculation uses explicit helicity spinors; the largest residual is 3.1×10^{-16} . Conjugation and equal-time anticommutators are checked as well. Raising special-function precision from 30 to 60 digits leaves the resulting double-precision matrices unchanged. These are free-propagator consistency checks, not an independent box integration. The independent eight-slot Wick enumerator gives 48 connected pairings and agrees with the cycle formula, and with half the Dirac result, for all 16 SK assignments at $\mu = 0.2, 0.73, 2$. Its test momenta are generic edge momenta: this is a local contraction test, not a momentum integral. The original-field implementation in `checks/aoki_weyl_checks.py` also compares both soft branches with an independent Bessel decomposition at $\mu = 0.2, 0.73, 2, 5$. It checks the four-type hard sums against Hankel scalar products, both mixed assignments, and the factorized spin cut for all 16 SK choices at the first three masses. The first-descendant remainder decreases quadratically with the soft momentum. These tests use the original $\psi, \psi^\dagger$ contractions. The soft-expansion check retains the exact ${}_0F_1$ endpoints, Whittaker hard lines, both external energy shifts and all four orderings. On a finite time interval, at three masses and two geometries, its error relative to the leading nonzero coefficient decreases by approximately four when the soft momenta are halved. The nonzero single-cycle zeroth-order trace and its exchange cancellation are checked separately. Time-quadrature refinement is also tested.

A further independent calculation integrates the fermion Hankel kernels on BD-rotated contours at the physical mass indices: the two same-SK ordered triangles rotate together, and the two opposite-SK integrals rotate separately. It checks both FF and GF scalar hard integrals and the hard-momentum derivative at $\mu = 0.2, 0.73$, with a common positive endpoint regulator $\delta = 0.8$, including a mixed-branch case. The derivative is obtained directly from the first-order mode equations, independently of the scalar bootstrap source; the regulated $\mathcal{P}_1, \mathcal{P}_2$ are compared as well. This tests the scalar-seed normalization beyond the conformal-scalar example. It does not directly evaluate the zero-regulator limit; that limit still uses the analytically summed seeds and regulator-path checks.

The retained scalar-seed tests check the elementary conformal-scalar normalization, Gamma reductions, regulator paths and working precision. Comparing the regulated and analytically summed versions of the same seed is a consistency check, not a second derivation of that seed. Neither finite-interval soft tests nor special-function agreement is a direct numerical integration of the full unexpanded BD box. The analytic contact part and its renormalization, overlapping collapsed channels, evolving masses, nonzero chemical potential and curvature conversion have not been computed. Reproducible commands and residuals are recorded in `checks/top_quark_audit.md`.

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